

Huao Li

Engineering Excellence Postdoctoral Fellow
Laboratory for Information and Decision Systems
Massachusetts Institute of Technology
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RESEARCH INTERESTS

Human-AI Teamwork: adaption, interpretable communication, trust calibration;
Multi-agent Systems: multi-agent reinforcement learning, embodied LLM agents;
Human-robot Interaction: human-in-the-loop robot learning, cognitive modeling;
Societal AI: explainability, value alignment, normative reasoning

EDUCATION

University of Pittsburgh

Ph.D. in Intelligent Systems

Pittsburgh, PA

Aug. 2019 – Aug. 2025

Thesis: A Computational Framework for Efficient Human-agent Teamwork

Committee members: Michael Lewis, Na Du, Daqing He, Katia Sycara (CMU)

University of Pittsburgh

Master of Science in Information Science

Pittsburgh, PA

Aug. 2017 – May 2019

Thesis: A Computational Model of Human Trust in Supervisory Control of Robotic Swarms

Zhejiang University

Bachelor of Science in Applied Psychology

Hangzhou, China

Sep. 2013 – June 2017

Thesis: The Influence of Pictorial Realism on the Comprehension of Safety Briefing Cards

RESEARCH EXPERIENCE

Massachusetts Institute of Technology

Engineering Excellence Postdoctoral Fellow, Advisor: Chuchu Fan

Cambridge, MA

Sep. 2025 – present

Focus: Socially intelligent LLM agents and Bayesian active learning for human-robot collaboration.

Carnegie Mellon University

Research Assistant, Advisors: Katia Sycara and Michael Lewis

Pittsburgh, PA

Aug. 2017 – Aug. 2025

Focus: Adaptive human-AI teamwork via deep reinforcement learning and theory of mind modeling.

Honda Research Institute USA

Research Scientist Intern

Ann Arbor, MI

Jan. 2024 – Aug. 2024

Focus: Language-grounded multi-agent RL for human-interpretable communication.

Zhejiang University

Research Assistant, Advisor: Zaifeng Gao

Hangzhou, China

Dec. 2015 – June 2017

Focus: Visual cognition in motion perception and human factors in driver distraction.

Alibaba Group

User Experience Researcher Intern

Hangzhou, China

Dec. 2015 – Aug. 2016

Focus: Cross-cultural user testing of AliExpress.com.

FELLOWSHIPS AND AWARDS

- | | |
|---|-----------|
| • MIT Postdoctoral Fellowship for Engineering Excellence | 2025-2027 |
| • Best Paper Award, <i>GenAI-HRI Workshop at Robotics: Science and Systems</i> | 2025 |
| • Best Paper Award, <i>AAAI Conference on Artificial Intelligence, Ethics, and Society (AIES)</i> | 2018 |

Book Chapter

- [B1] Michael Lewis, **Huao Li**, and Katia Sycara (2021). [Deep learning, transparency, and trust in human robot teamwork](#). *Trust in Human-Robot Interaction*.

Peer Reviewed Journal Papers

- [J1] Lesong Jia, Kiran Shridhar Alva, **Huao Li**, Werner Hager, Lucia Romero, Timothy Albert Butler, and Michael Lewis (2026). [Understanding Situation Awareness in Multi-UAV Supervision: Effects of Environmental Complexity, Interface Design, and Human–AI Differences](#). *International Journal of Human–Computer Interaction (IJHCI)*.
- [J2] Seth Karten, Mycal Tucker, **Huao Li**, Siva Kailas, Michael Lewis, and Katia Sycara (2023). [Interpretable Learned Emergent Communication for Human–Agent Teams](#). *IEEE Transactions on Cognitive and Developmental Systems (IEEE T-CDS)*.
- [J3] **Huao Li**, Tianwei Ni, Siddharth Agrawal, Fan Jia, Suhas Raja, Yikang Gui, Dana Hughes, Michael Lewis, and Katia Sycara (2021). [Individualized Mutual Adaptation in Human-Agent Teams](#). *IEEE Transactions on Human-Machine Systems (IEEE T-HMS)*.
- [J4] Changjoo Nam, Phillip Walker, **Huao Li**, Michael Lewis, and Katia Sycara (2020). [Models of trust in human control of swarms with varied levels of autonomy](#). *IEEE Transactions on Human-Machine Systems (IEEE T-HMS)*. **121 citations**.
- [J5] Tian-mi Fang, **Huao Li**, Zheng Ma, Ze-ya Peng, Yi-ping Xie, Cheng-feng Zhu, Ci Wang, Chao Song, and Zai-feng Gao (2017). [The Influence of Pictorial Realism on the Comprehension of Safety Briefing Cards](#). *Chinese Journal of Applied Psychology (Chin. J. Appl. Psychol.)* Corresponding author.

Peer Reviewed Conference Papers

- [C1] **Huao Li**, Carson Sobolewski, Augustinos D Saravanos, Tan Yew Teck, John Karigiannis, and Chuchu Fan (2026). [Bayesian Active Learning for Intent Disambiguation in Interactive Robot Planning](#). *Conference on Robot Learning (CoRL)*. Accepted.
- [C2] Benjamin Li, Shuyang Shi, Lucia Romero, **Huao Li**, Yaqi Xie, Woojun Kim, Stefanos Nikolaidis, Charles Lewis, Katia Sycara, and Simon Stepputtis (2025). [Adaptively coordinating with novel partners via learned latent strategies](#). *Conference on Neural Information Processing Systems (NeurIPS)*.
- [C3] **Huao Li**, Hossein Nourkhiz Mahjoub, Behdad Chalaki, Vaishnav Tadiparthi, Kwonjoon Lee, Ehsan Moradi-Pari, Charles Michael Lewis, and Katia P. Sycara (2024). [Language Grounded Multi-agent Reinforcement Learning with Human-interpretable Communication](#). *Conference on Neural Information Processing Systems (NeurIPS)*. **51 citations**.
- [C4] Xuehang Guo, **Huao Li**, Anfeng Peng, Lesong Jia, Brandon Rishi, and Michael Lewis (2024). [Determining human perceptual envelope in fixed wing UAV surveillance](#). *IEEE International Conference on Human-Machine Systems (ICHMS)*.
- [C5] Lesong Jia, Anfeng Peng, **Huao Li**, Xuehang Guo, and Michael Lewis (2024). [Situation Theory Based Query Generation for Determining Situation Awareness Across Distributed Data Streams](#). *IEEE International Conference on Human-Machine Systems (ICHMS)*.
- [C6] **Huao Li**, Yu Quan Chong, Simon Stepputtis, Joseph Campbell, Dana Hughes, Michael Lewis, and Katia Sycara (2023). [Theory of Mind for Multi-Agent Collaboration via Large Language Models](#). *Conference on Empirical Methods in Natural Language Processing (EMNLP)*. **373 citations**.
- [C7] **Huao Li**, Yao Fan, Keyang Zheng, Michael Lewis, and Katia Sycara (2023). [Personalized Decision Supports based on Theory of Mind Modeling and Explainable Reinforcement Learning](#). *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*.
- [C8] Seth Karten, Siva Kailas, **Huao Li**, and Katia Sycara (2023). [On the Role of Emergent Communication for Social Learning in Multi-Agent Reinforcement Learning](#). *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*.
- [C9] Dana Hughes, **Huao Li**, Max Chis, Ini Oguntola, Simon Stepputtis, Keyang Zheng, Joseph Campbell, Katia Sycara, and Michael Lewis (2023). [A Framework for Intervention Based Team Support in Time Critical Tasks](#). *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*.
- [C10] Maximillian Chis, **Huao Li**, Keyang Zheng, Michael Lewis, Dana Hughes, and Katia Sycara (2023). [The Cognitive Load–Productivity Tradeoff in Task Switching](#). *Human Factors and Ergonomics Society Annual Meeting (HFES)*.

- [C11] **Huao Li**, Maximillian Chis, Keyang Zheng, Michael Lewis, Dana Hughes, and Katia Sycara (2023). Sentiment Analysis of Artificial Advisors in Search and Rescue Tasks. *Human Factors and Ergonomics Society Annual Meeting (HFES)*.
- [C12] **Huao Li**, Ini Oguntola, Dana Hughes, Michael Lewis, and Katia Sycara (2022). [Theory of Mind Modeling in Search and Rescue Teams](#). *IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*.
- [C13] Ankur Deka, Katia Sycara, Phillip Walker, **Huao Li**, and Michael Lewis (2022). The robustness of human advantage in swarm leader identification. *Human Factors and Ergonomics Society Annual Meeting (HFES)*.
- [C14] Mycal Tucker, **Huao Li**, Siddharth Agrawal, Dana Hughes, Katia Sycara, Michael Lewis, and Julie A. Shah (2021). [Emergent Discrete Communication in Semantic Spaces](#). *Conference on Neural Information Processing Systems (NeurIPS)*. **54 citations**.
- [C15] Ankur Deka, Wenhao Luo, **Huao Li**, Michael Lewis, and Katia Sycara (2021). [Hiding Leader's Identity in Leader-Follower Navigation through Multi-Agent Reinforcement Learning](#). *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.
- [C16] **Huao Li**, Keyang Zheng, Michael Lewis, Dana Hughes, and Katia Sycara (2021). [Human Theory of Mind Inference in Search and Rescue Tasks](#). *Human Factors and Ergonomics Society Annual Meeting (HFES)*.
- [C17] Ankur Deka, Katia Sycara, Phillip Walker, **Huao Li**, and Michael Lewis (2021). [Human vs. deep neural network performance at a leader identification task](#). *Human Factors and Ergonomics Society Annual Meeting (HFES)*.
- [C18] **Huao Li**, Michael Lewis, and Katia Sycara (2020). [A Kalman estimation model of human trust in supervisory control of robotic swarms](#). *Human Factors and Ergonomics Society Annual Meeting (HFES)*.
- [C19] **Huao Li**, Tianwei Ni, Siddharth Agrawal, Dana Hughes, and Katia Sycara (2020). [Team Synchronization and Individual Contributions in Coop-Space Fortress](#). *Human Factors and Ergonomics Society Annual Meeting (HFES)*.
- [C20] Katia Sycara, Dana Hughes, **Huao Li**, Michael Lewis, and Nina Lauharatanahirun (2020). Adaptation in Human-Autonomy Teamwork. *IEEE International Conference on Human-Machine Systems (ICHMS)*.
- [C21] **Huao Li**, Stephanie Milani, Vigneshram Krishnamoorthy, Michael Lewis, and Katia Sycara (2019). [Perceptions of domestic robots' normative behavior across cultures](#). *AAAI/ACM Conference on AI, Ethics, and Society (AIES)*.
- [C22] **Huao Li**, Jaeho Bang, Sasanka Nagavalli, Changjoo Nam, Michael Lewis, and Katia Sycara (2018). [Human Interaction Through an Optimal Sequencer to Control Robotic Swarms](#). *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*.
- [C23] Rahul Iyer, Yuezhang Li, **Huao Li**, Michael Lewis, Ramitha Sundar, and Katia Sycara (2018). [Transparency and explanation in deep reinforcement learning neural networks](#). *AAAI/ACM Conference on AI, Ethics, and Society (AIES)*. **Best Paper Award**. **256 citations**.
- [C24] Changjoo Nam, **Huao Li**, Shen Li, Michael Lewis, and Katia Sycara (2018). [Trust of humans in supervisory control of swarm robots with varied levels of autonomy](#). *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*.

Workshop Papers and Abstracts

- [W1] Benjamin Li, Shuyang Shi, Lucia Romero, **Huao Li**, Yaqi Xie, Woojun Kim, Stefanos Nikolaidis, Michael Lewis, Katia Sycara, and Simon Stepputtis (2025). [Modeling Latent Partner Strategies for Adaptive Zero-Shot Human-Agent Collaboration](#). *Generative AI for Human-Robot Interaction Workshop at Robotics: Science and Systems (GenAI-HRI at RSS)*. **Best Paper Award**.
- [W2] **Huao Li**, Long Le, Max Chis, Keyang Zheng, Dana Hughes, Michael Lewis, and Katia Sycara (2021). [Sequential theory of mind modeling in team search and rescue tasks](#). *AAAI Fall Symposium on Computational Theory of Mind for Human-Machine Teams (AAAI Symposium)*.
- [W3] **Huao Li**, Dana Hughes, Michael Lewis, and Katia Sycara (2020). [Individual adaptation in teamwork](#). *Annual Meeting of the Cognitive Science Society (CogSci)*. Abstract.
- [W4] Vidhi Jain, Rohit Jena, **Huao Li**, Tejus Gupta, Dana Hughes, Michael Lewis, and Katia Sycara (2020). [Predicting human strategies in simulated search and rescue task](#). *Artificial Intelligence for Humanitarian Assistance and Disaster Response Workshop at Neural Information Processing Systems (AI-HADR at NeurIPS)*.

MENTORING

Leonardo Paschoal Bartoccini

High school student

Research Topic: Value-based intrinsic rewards for cooperation in sequential social dilemmas

MIT RSI Program

Summer 2026

Dami Thomas

Undergraduate student at MIT

Research Topic: Safe multi-agent control for autonomous field inspection robots

MIT UROP

Fall 2025

Owen Burns

Undergraduate student at University of Central Florida

Research Topic: Human-in-the-loop formal planning and symbolic plan alignment

CMU RISS Program

Summer 2024

Lucia Romero

MSIS student at University of Pittsburgh

Research Topic: A benchmark for human-agent collaboration in Overcooked

Resulted in 1 AAAI '27 submission.

University of Pittsburgh

2024–2025

Xuehang Guo

MSIS student at University of Pittsburgh

Research Topic: Situation awareness in multi-UAV supervision

Resulted in 1 IEEE ICHMS '24 publication.

University of Pittsburgh

2023–2024

Max Chis

MSIS student at University of Pittsburgh

Research Topic: Team interventions in time-critical search and rescue

Resulted in 1 HFES '23 publication.

University of Pittsburgh

2022–2023

Seth Karten

MS student in Robotics at Carnegie Mellon University

Research Topic: Emergent communication for social learning in agent teams

Resulted in 1 AAMAS '23 and 1 IEEE T-CDS '23 publication.

Carnegie Mellon University

2021–2023

Long Le

Undergraduate student at UMass Amherst

Research Topic: Theory of mind and multi-agent RL for team search and rescue

Resulted in 1 AAAI Symposium '21 publication.

CMU RISS Program

Summer 2021

Suhas Raja

Undergraduate student at UT Austin

Research Topic: Adaptive agents for individualized human-agent teamwork

Resulted in 1 IEEE T-HMS '21 publication.

CMU RISS Program

Summer 2020

Stephanie Milani

Undergraduate student at UMBC

Research Topic: Reinforcement learning in norm-rich environments

Resulted in 1 AIES '19 publication.

CMU RISS Program

Summer 2018

TEACHING

Kaufman Teaching Certificate Program

Participant

MIT

Spring 2026

INFSCI 2300 Human Information Processing

Teaching Assistant

University of Pittsburgh

Fall 2024

INFSCI 2935 Human Robot Interaction

Teaching Assistant

University of Pittsburgh

Fall 2022, Fall 2023

FUNDED PROJECTS

Autonomous Field Inspection with Physical AI

Co-I, PI: Chuchu Fan (MIT)

GE Vernova

2025/09–2027/09

Exploratory Models of Human-AI Teams

Co-I, PIs: Katia Sycara (CMU), Stefanos Nikolaidis (USC)

DARPA

2024/04–2026/04

Robust Collaborative AI

Co-I, PIs: Katia Sycara (CMU), Sherry Tongshuang Wu (CMU)

ONR

2023/06–2027/06

A Robust and Adaptive Agent that Support High Performance Teams

Co-I, PIs: Katia Sycara (CMU), Michael Lewis (Pitt)

DARPA

2019/06–2023/05

Strengthening Teamwork for Robust Operations in Novel Groups

Co-I, PIs: Katia Sycara (CMU), Michael Lewis (Pitt), Julie Shah (MIT)

ARL

2019/10–2022/09

Trustworthy Human Interaction with Robotic Swarms

Co-I, PIs: Katia Sycara (CMU), Michael Lewis (Pitt)

AFOSR

2018/10–2021/09

PROFESSIONAL SERVICES

- *MIT LIDS Student Conference 2026* Co-chair
- *MIT LIDS Postdoc Nexus Meeting* Committee
- SCI Faculty search committee, University of Pittsburgh Student Representative
- *Behaviour & Information Technology, Scientific Reports, IJCHI, T-HMS, PLOS One* Journal Reviewer
- *ICLR, NeurIPS, AAAI, EMNLP, ICRA, IROS, SMC, HFES* Conference Reviewer

REFERENCES

Prof. Chuchu Fan

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Cambridge, MA 02139, USA

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Prof. Michael Lewis

University of Pittsburgh
135 N. Bellefield Avenue, Room 2A00
Pittsburgh, PA 15260, USA

ml@sis.pitt.edu

+1 (412) 624-9426

Prof. Katia Sycara

Carnegie Mellon University
5000 Forbes Avenue, Newell-Simon Hall 1602D
Pittsburgh, PA 15213-3890, USA

sycara@andrew.cmu.edu

+1 (412) 268-8825

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70 Rio Robles
San Jose, CA 95134, USA

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